9.31 **Fig. P9.31** depicts a half-wave dipole connected to a generator through a matched transmission line. The directivity of the dipole can be modified by placing a reflecting rod a distance d behind the dipole. What would its reflectivity in the forward direction be if

(a) $d = \lambda/4$
(b) $d = \lambda/2$

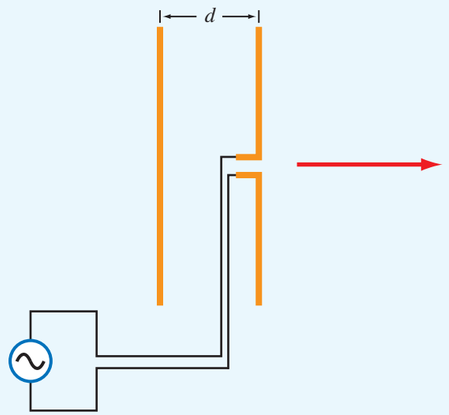


Figure P9.31 Problem 9.31.

9.32 The configuration shown in **Fig. P9.32** depicts a satellite repeater with two antennas, one pointed towards the antenna of ground station 1 and the other towards the antenna of ground station 2. All antennas are parabolic dishes; antennas A_1 and A_4 are each 4 m in diameter, antennas A_2 and A_3 are each 2 m in diameter, and the distance between the satellite and each of the ground stations is 40,000 km. Upon receiving the signal by its antenna A_2, the satellite transponder boosts the power gain by 80 dB and then retransmits the signal to A_4. The system operates at 10 GHz with $P_t = 1$ kW. Determine the received power P_r. Assume all antennas to be lossless.

Sections 9-7 and 9-8: Radiation by Apertures

*9.33 A uniformly illuminated aperture is of length $l_x = 20\lambda$. Determine the beamwidth between first nulls in the x–z plane.

9.34 The 10 dB beamwidth is the beam size between the angles at which $F(\theta)$ is 10 dB below its peak value. Determine the 10 dB beamwidth in the x–z plane for a uniformly illuminated aperture with length $l_x = 10\lambda$.

*9.35 A uniformly illuminated rectangular aperture situated in the x–y plane is 2 m high (along x) and 1 m wide (along y). If $f = 10$ GHz, determine:

(a) the beamwidths of the radiation pattern in the elevation plane (x–z plane) and the azimuth plane (y–z plane), and
(b) the antenna directivity D in decibels.

9.36 An antenna with a circular aperture has a circular beam with a beamwidth of 3° at 20 GHz.

(a) What is the antenna directivity in dB?
(b) If the antenna area is doubled, what will be the new directivity and new beamwidth?
(c) If the aperture is kept the same as in (a), but the frequency is doubled to 40 GHz, what will the directivity and beamwidth become then?

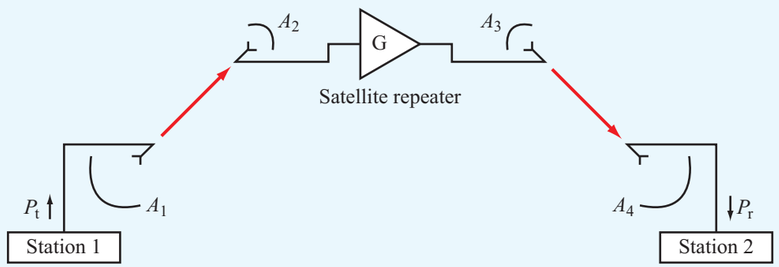


Figure P9.32 Satellite repeater system.

9.31 图 P9.31 描绘了通过匹配传输线连接到发电机的半波偶极子。偶极子的方向性可以通过在偶极子后面距离 d 处放置一根反射棒来改变。如果它向前的反射率是多少

(a) $d = \lambda/4$
(b) $d = \lambda/2$

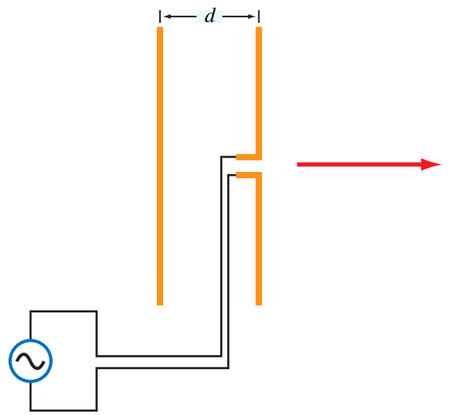


图 P9.31 问题 9.31。

9.32 图 P9.32 所示的配置描述了一个带有两根天线的卫星中继器，一根指向地面站 1 的天线，另一根指向地面站 2 的天线。所有天线都是抛物面天线；天线 A_1 和 A_4 的直径均为 4 m，天线 A_2 和 A_3 的直径为每个卫星直径为2 m，卫星与每个地面站的距离为4万公里。卫星转发器通过天线 A_2 接收到信号后，将功率增益提高 80 dB，然后将信号重新发送到 A_4。该系统工作频率为 10 GHz，$P_t = 1$ kW。确定接收功率 P_r。假设所有天线都是无损的。

第 9-7 节和 9-8 节：孔径辐射

*9.33 均匀照明孔径的长度为 $l_x = 20\lambda$。确定 x-z 平面中第一个零点之间的波束宽度。

9.34 10 dB 波束宽度是 F(θ) 低于其峰值 10 dB 的角度之间的波束大小。确定长度为 $l_x = 10\lambda$ 的均匀照明孔径在 x-z 平面上的 10 dB 波束宽度。

*9.35 位于 x-y 平面上的均匀照明矩形孔径为 2 m 高（沿 x）和 1 m 宽（沿 y）。如果 f = 10 GHz，则确定：

(a) 仰角平面（x-z 平面）和方位角平面（y-z 平面）中辐射方向图的波束宽度，以及 (b) 天线方向性 D（以分贝为单位）。

9.36 圆形孔径天线的圆形波束在 20 GHz 时波束宽度为 3°。

(a) 天线方向性（以 dB 为单位）是多少？ (b) 如果天线面积加倍，新的方向性和新的波束宽度将是多少？ (c) 如果孔径保持与 (a) 相同，但频率加倍至 40 GHz，那么方向性和波束宽度会变成什么？

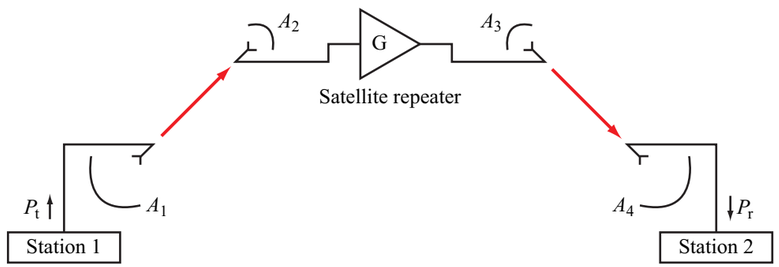


图 P9.32 卫星中继器系统。

9.37 Compare directivity D_{ant} of a 1-m diameter antenna aperture operating at 10 GHz with directivity D_{eye} of the eye's pupil operating in the middle of the visible spectrum at $\lambda = 0.5\ \mu$m. Treat the pupil as a circular aperture with a diameter of 4 mm.

*9.38 A 94 GHz automobile collision-avoidance radar uses a rectangular-aperture antenna placed above the car's bumper. If the antenna is 1 m in length and 10 cm in height, determine:

(a) its elevation and azimuth beamwidths, and

(b) the horizontal extent of the beam at a distance of 300 m.

9.39 A microwave telescope consisting of a very sensitive receiver connected to a 100 m parabolic-dish antenna is used to measure the energy radiated by astronomical objects at 20 GHz. If the antenna beam is directed toward the moon and the moon extends over a planar angle of 0.5° from Earth, what fraction of the moon's cross-section will be occupied by the beam?

9.40 A parabolic dish antenna has a diameter $d = 5\lambda$. Use Module 9.4 to evaluate three antenna parameters, namely the directivity D, the beamwidth β, and the level of the first sidelobe next to the mainbeam, all at the following three values of the taper factor: $\alpha = 0$, $\alpha = 0.5$, and $\alpha = 1$.

9.41 Given a parabolic dish antenna with a diameter $d = 10\lambda$, use Module 9.4 to select the highest value of the taper factor α without having the beamwidth exceed 6°.

Sections 9-9 and 9-11: Antenna Arrays

9.42 A two-element array consisting of two isotropic antennas separated by a distance d along the z axis is placed in a coordinate system whose z axis points eastward and whose x axis points toward the zenith. If a_0 and a_1 are the amplitudes of the excitations of the antennas at $z = 0$ and at $z = d$, respectively, and if δ is the phase of the excitation of the antenna at $z = d$ relative to that of the other antenna, find the array factor and plot the pattern in the x–z plane for the following.

*(a) $a_0 = a_1 = 1,\ \delta = \pi/4$, and $d = \lambda/2$

(b) $a_0 = 1,\ a_1 = 2,\ \delta = 0$, and $d = \lambda$

(c) $a_0 = a_1 = 1,\ \delta = -\pi/2$, and $d = \lambda/2$

(d) $a_0 = 1,\ a_1 = 2,\ \delta = \pi/4$, and $d = \lambda/2$

(e) $a_0 = 1,\ a_1 = 2,\ \delta = \pi/2$, and $d = \lambda/4$

9.43 If the antennas in part (a) of Problem 9.42 are parallel, vertical, Hertzian dipoles with axes along the x direction, determine the normalized radiation intensity in the x–z plane and plot it.

*9.44 Consider the two-element dipole array of **Fig. 9-29(a)**. If the two dipoles are excited with identical feeding coefficients ($a_0 = a_1 = 1$ and $\psi_0 = \psi_1 = 0$), choose (d/λ) so that the array factor has a maximum at $\theta = 45°$.

9.45 Choose (d/λ) so that the array pattern of the array of Problem 9.44 has a null, rather than a maximum, at $\theta = 45°$.

9.46 Find and plot the normalized array factor and determine the half-power beamwidth for a five-element linear array excited with equal phase and a uniform amplitude distribution. The interelement spacing is $3\lambda/4$.

9.47 Repeat Problem 9.46, but change the excitation to tapered amplitude distribution so that the amplitude of the central element is 1, the amplitudes of the next adjacent elements are both 0.5, and those of the outer elements are both 0.25.

9.48 Repeat Problem 9.46 for a nine-element array.

*9.49 A five-element equally spaced linear array with $d = \lambda/2$ is excited with uniform phase and an amplitude distribution given by the binomial distribution

$$a_i = \frac{(N-1)!}{i!(N-i-1)!}, \qquad i = 0, 1, \ldots, (N-1),$$

where N is the number of elements. Develop an expression for the array factor.

9.50 A three-element linear array of isotropic sources aligned along the z axis has an interelement spacing of $\lambda/4$ (**Fig. P9.50**). The amplitude excitation of the center element is twice that of the bottom and top elements, and the phases are $-\pi/2$ for the bottom element and $\pi/2$ for the top element relative to that of the center element. Determine the array factor and plot it in the elevation plane.

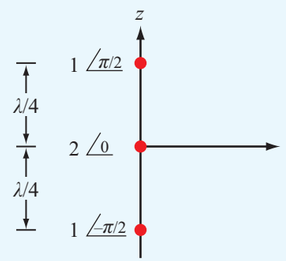


Figure P9.50 Three-element array of Problem 9.50.

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9.37 比较在 10 GHz 工作的 1 米直径天线孔径的方向性 D_{ant} 与在可见光谱中间工作在 $\lambda = 0.5\ \mu$m 的眼睛瞳孔的方向性 D_{eye}。将瞳孔视为直径为 4 mm 的圆形光圈。

*9.38 94 GHz 汽车防撞雷达使用放置在汽车保险杠上方的矩形孔径天线。如果天线长1 m，高10 cm，则求：

(a) 其仰角和方位角波束宽度，以及 (b) 300 m 距离处波束的水平范围。

9.39 微波望远镜由连接到 100 m 抛物面天线的非常灵敏的接收器组成，用于测量天体在 20 GHz 辐射的能量。如果天线波束指向月球，并且月球与地球的平面角为 0.5°，则波束将占据月球横截面的几分之一？

9.40 抛物面碟形天线的直径为 $d = 5\lambda$。使用模块 9.4 评估三个天线参数，即方向性 D、波束宽度 β 和主波束旁边的第一个旁瓣的电平，所有参数均采用以下三个锥度因子值：$\alpha = 0$、$\alpha = 0.5$ 和 $\alpha = 1$。

9.41 给定直径为 $d = 10\lambda$ 的抛物面天线，使用模块 9.4 选择锥度系数 α 的最高值，且波束宽度不超过 6°。

第 9-9 和 9-11 节：天线阵列

9.42 由沿 z 轴间隔距离 d 的两个各向同性天线组成的二元阵列放置在 z 轴指向东且 x 轴指向天顶的坐标系中。如果 a_0 和 a_1 分别是天线在 $z = 0$ 和 $z = d$ 处的激励幅度，并且如果 δ 是天线在 $z = d$ 处相对于另一个天线的激励相位，则求出阵列因子并在 x-z 平面上绘制以下方向图。

*(a) $a_0 = a_1 = 1$、$\delta = \pi/4$ 和 $d = \lambda/2$ (b) $a_0 = 1$、$a_1 = 2$、$\delta = 0$ 和 $d = \lambda$ c) $a_0 = a_1 = 1$、$\delta = -\pi/2$ 和 $d = \lambda/2$ (d) $a_0 = 1$、$a_1 = 2$、$\delta = \pi/4$ 和 $d = \lambda/2$ (e) $a_0 = 1$、$a_1 = 2$、$\delta = \pi/2$ 和 $d = \lambda/4$

9.43 如果问题 9.42 (a) 部分中的天线是平行、垂直、轴沿 x 方向的赫兹偶极子，则确定 x-z 平面中的归一化辐射强度并将其绘制出来。

*9.44 考虑图 9-29(a) 的二元偶极子阵列。如果两个偶极子以相同的馈电系数（$a_0 = a_1 = 1$ 和 $\psi_0 = \psi_1 = 0$）激发，请选择（d/λ），以便阵列因子在 $\theta = 45°$ 处具有最大值。

9.45 选择(d/λ)，使得问题9.44的数组的数组模式在$\theta = 45°$处具有空值，而不是最大值。

9.46 求并绘制归一化阵列因子，并确定以等相位和均匀幅度分布激励的五元线性阵列的半功率波束宽度。元素间间距为 $3\lambda/4$。

9.47 重复问题9.46，但将激励改为锥形振幅分布，使得中心单元的振幅为1，接下来的相邻单元的振幅均为0.5，外部单元的振幅均为0.25。

9.48 对九元素数组重复问题 9.46。

*9.49 具有 $d = \lambda/2$ 的五元等距线性阵列由均匀相位和由二项式分布给出的幅度分布激励

$$a_i = \frac{(N-1)!}{i!(N-i-1)!}, \qquad i = 0, 1, \ldots, (N-1),$$

其中 N 是元素的数量。开发数组因子的表达式。

9.50 沿 z 轴排列的各向同性源的三元线性阵列的元间间距为 $\lambda/4$（图 P9.50）。中心单元的振幅激励是底部和顶部单元的两倍，相对于中心单元的相位，底部单元的相位为$-\pi/2$，顶部单元的相位为$\pi/2$。确定阵列系数并将其绘制在高程平面上。

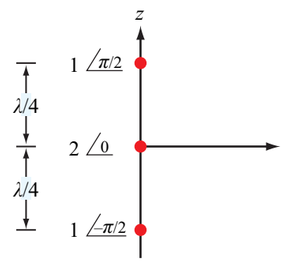


图 P9.50 问题 9.50 的三元素数组。

*9.51 An eight-element linear array with $\lambda/2$ spacing is excited with equal amplitudes. To steer the main beam to a direction 60° below the broadside direction, what should be the incremental phase delay between adjacent elements? Also, give the expression for the array factor and plot the pattern.

9.52 A linear array arranged along the z axis consists of 12 equally spaced elements with $d = \lambda/2$. Choose an appropriate incremental phase delay δ so as to steer the main beam to a direction 30° above the broadside direction. Provide an expression for the array factor of the steered antenna and plot the pattern. From the pattern, estimate the beamwidth.

9.53 Use Module 9.5 to obtain the radiation pattern for a two-dipole array, each 0.5λ in length, separated by $\lambda/2$, and fed equally in amplitude but with one of them 90° ahead of the other in phase.

9.54 Repeat Problem 9.53, but change the separation between the two dipoles to $d = 1\lambda$.

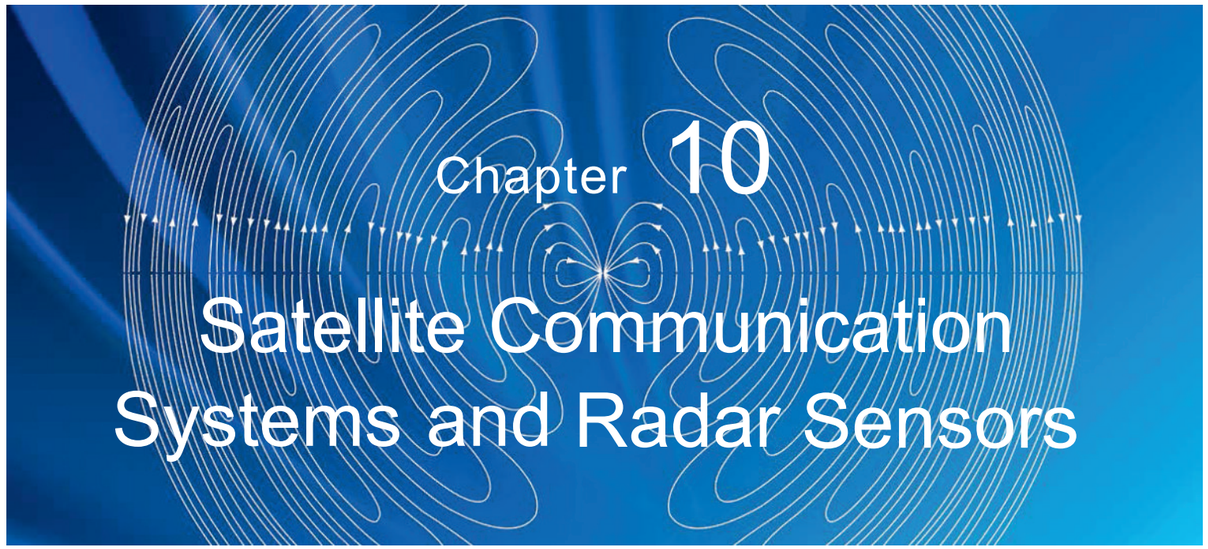
PROBLEMS 433

*9.51 具有λ/2 间距的八元线性阵列以等幅激励。为了将主波束转向侧射方向下方 60° 的方向，相邻元件之间的增量相位延迟应该是多少？另外，给出数组因子的表达式并绘制模式。

9.52 沿 z 轴排列的线性阵列由 12 个等间隔的元素组成，元素的间距为 d = λ/2。选择适当的增量相位延迟 δ，以便将主波束转向高于侧射方向 30° 的方向。提供一个可控天线阵列因子的表达式并绘制方向图。根据该图案，估计波束宽度。

9.53 使用模块 9.5 获得双偶极子阵列的辐射方向图，每个偶极子阵列长度为 0.5λ，间隔为 λ/2，并且馈电幅度相等，但其中一个在相位上领先另一个 90°。

9.54 重复问题 9.53，但将两个偶极子之间的间隔更改为 d = 1λ。



Chapter 10
Satellite Communication Systems and Radar Sensors

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Objectives

Upon learning the material presented in this chapter, you should be able to:

1. Describe the basic operation of satellite transponders.
2. Calculate the power budget for a communication link.
3. Describe how radar attains spatial and angular resolutions, calculate the maximum detectable range, and explain the trade-off between the probabilities of detection and false alarm.
4. Calculate the Doppler frequency shift observed by a radar.
5. Describe the monopulse-radar technique.

第10章

卫星通信系统与雷达传感器

章节内容

应用示例、435 10-1 卫星通信系统、435 10-2 卫星转发器、436 10-3 通信链路功率预算、439 10-4 天线波束、440 10-5 雷达传感器、441 10-6 目标检测、443 10-7 多普勒雷达、445 10-8 单脉冲雷达、 447 第 10 章总结，449 问题，450

目标

学习本章介绍的材料后，您应该能够：

1. 描述卫星转发器的基本操作。 2. 计算通信链路的功率预算。 3. 描述雷达如何获得空间和角度分辨率，计算最大可检测范围，并解释检测概率和误报概率之间的权衡。 4. 计算雷达观测到的多普勒频移。 5. 描述单脉冲雷达技术。

Application Examples

This concluding chapter presents overviews of satellite communication systems and radar sensors with emphasis on their electromagnetic-related aspects.

10-1 Satellite Communication Systems

Today's world is connected by a vast communication network that provides a wide array of voice, data, and video services to both fixed and mobile terminals (**Fig. 10-1**). The viability and effectiveness of the network are attributed in large measure to the use of orbiting satellite systems that function as relay stations with wide area coverage of Earth's surface. From a geostationary orbit at 35,786 km above the equator, a satellite can view over one-third of Earth's surface and can connect any pair of points within its coverage (**Fig. 10-2**). The history of communication satellite engineering dates back to the late 1950s when the U.S. Navy used the moon as a passive reflector to relay low-data-rate communications between Washington D.C. and Hawaii. The first major development involving artificial Earth satellites took place in October of 1957 when the Soviet Union launched ***Sputnik I*** and used it for 21 days to transmit (one-way) telemetry information to a ground receiving station. This was followed by another telemetry satellite, ***Explorer I***, launched by the United States in January, 1958. An important development took place in December of that year when the United States launched the ***Score*** satellite and used it to broadcast President Eisenhower's Christmas message, marking the first instance of two-way voice communication via an artificial satellite.

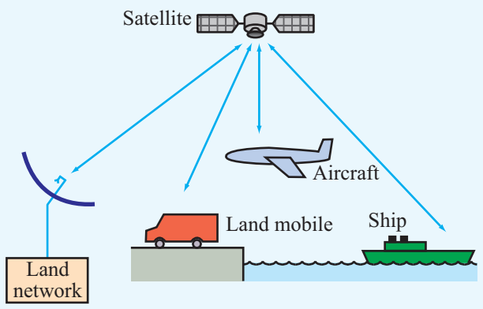
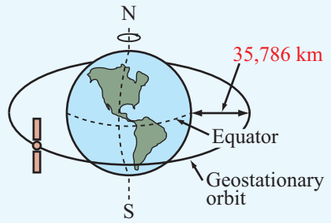


Figure 10-1 Elements of a satellite communication network.

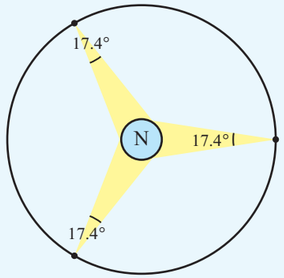
These achievements were followed by a flurry of space activity, leading to the development of operational communication satellites by many countries for both commercial and governmental services. This section describes satellite communications links with emphasis on transmitter–receiver power calculations, propagation aspects, frequency allocations, and antenna design considerations.

A satellite is said to be in a ***geostationary orbit*** around the Earth when it is in a circular orbit in a plane identical with Earth's equatorial plane at an altitude where the orbital period is identical with Earth's rotational period, thereby appearing stationary relative to Earth's surface. A satellite of mass M_s in circular orbit around Earth (**Fig. 10-3**) is subject to two forces: the attractive gravitational force F_g and the repelling centrifugal force F_c. The magnitudes of these two forces are given by

$$F_g = \frac{GM_sM_e}{R_0^2}, \tag{10.1}$$



(a) Geostationary satellite orbit



(b) Worldwide coverage by three satellites spaced 120° apart

Figure 10-2 Orbits of geostationary satellites.

应用实例

最后一章概述了卫星通信系统和雷达传感器，重点介绍了它们的电磁相关方面。

10-1 卫星通信系统

当今世界通过庞大的通信网络连接起来，为固定和移动终端提供广泛的语音、数据和视频服务（图10-1）。该网络的可行性和有效性在很大程度上归功于轨道卫星系统的使用，这些系统充当中继站，广泛覆盖地球表面。从赤道上空 35,786 公里的地球静止轨道上，卫星可以看到地球表面的三分之一以上，并可以连接其覆盖范围内的任意一对点（图 10-2）。通信卫星工程的历史可以追溯到 20 世纪 50 年代末，当时美国海军利用月球作为无源反射器来中继华盛顿特区和夏威夷之间的低数据速率通信。涉及人造地球卫星的第一个重大发展发生在 1957 年 10 月，当时苏联发射了 Sputnik I，并使用它长达 21 天向地面接收站传输（单向）遥测信息。随后，美国于 1958 年 1 月发射了另一颗遥测卫星 Explorer I。同年 12 月，美国发射了 Score 卫星，并用它来广播艾森豪威尔总统的圣诞致辞，这是一项重要的进展，标志　着通过人造卫星进行双向语音通信的首次实例。

这些成就之后是一系列太空活动，导致许多国家开发了用于商业和政府服务的业务通信卫星。本节描述卫星通信链路，重点是发射机-接收机功率计算、传播方面、频率分配和天线设计注意事项。

当卫星处于与地球赤道平面相同的平面上的圆形轨道上且轨道周期与地球自转周期相同的高度时，卫星被称为处于绕地球地球静止轨道，因此相对于地球表面而言是静止的。质量为 M_s 的卫星在绕地球运行的圆形轨道上（图 10-3）受到两种力的作用：吸引力 F_g 和排斥离心力 F_c。这两个力的大小由下式给出

$$F_g = \frac{GM_sM_e}{R_0^2}, \tag{10.1}$$

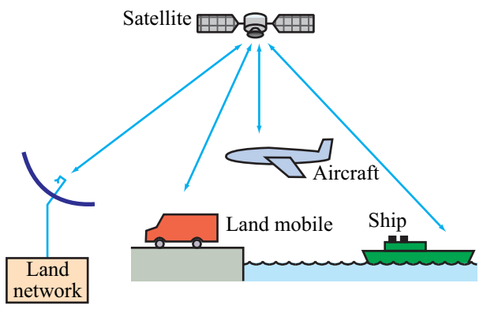
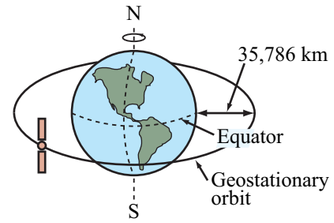
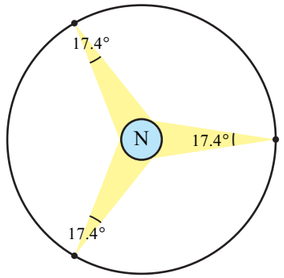


图 10-1 卫星通信网络的要素。



(a) Geostationary satellite orbit



(b) Worldwide coverage by three satellites spaced 120° apart

图10-2 对地静止卫星的轨道。